An Example Database

Transaction database

Tid	Itemset
1	$ABDE$
2	BCE
3	$ABDE$
4	$ABCE$
5	$ABCDE$
6	BCD

Frequent itemsets ($minsup = 3$)

sup	Itemsets
6	B
5	E, BE
4	$A, C, D, AB, AE, BC, BD, ABE$
3	$AD, CE, DE, ABD, ADE, BCE, BDE, ABDE$

Maximal Frequent Itemsets

Given a binary database $\mathbf{D} \subseteq \mathcal{T} \times \mathcal{I}$, over the tids $\mathcal{T}$ and items $\mathcal{I}$, let $\mathcal{F}$ denote the set of all frequent itemsets, that is,

$$\mathcal{F} = \left\{ X \mid X \subseteq \mathcal{I} \text{ and } sup(X) \geq minsup \right\}$$

A frequent itemset $X \in \mathcal{F}$ is called *maximal* if it has no frequent supersets. Let $\mathcal{M}$ be the set of all maximal frequent itemsets, given as

$$\mathcal{M} = \left\{ X \mid X \in \mathcal{F} \text{ and } \nexists Y \supset X, \text{ such that } Y \in \mathcal{F} \right\}$$

The set $\mathcal{M}$ is a condensed representation of the set of all frequent itemset $\mathcal{F}$, because we can determine whether any itemset X is frequent or not using $\mathcal{M}$. If there exists a maximal itemset Z such that $X \subseteq Z$, then X must be frequent; otherwise X cannot be frequent.

Closed Frequent Itemsets

Given $T \subseteq \mathcal{T}$, and $X \subseteq \mathcal{I}$, define

$$\begin{aligned} \boldsymbol{t}(X) &= \{t \in \mathcal{T} \mid t \text{ contains } X\} \\ \boldsymbol{i}(T) &= \{x \in \mathcal{I} \mid \forall t \in T,\ t \text{ contains } x\} \\ \boldsymbol{c}(X) &= \boldsymbol{i} \circ \boldsymbol{t}(X) = \boldsymbol{i}(\boldsymbol{t}(X)) \end{aligned}$$

The function $\boldsymbol{c}$ is a *closure operator* and an itemset X is called *closed* if $\boldsymbol{c}(X) = X$. It follows that $\boldsymbol{t}(\boldsymbol{c}(X)) = \boldsymbol{t}(X)$. The set of all closed frequent itemsets is thus defined as

$$\boxed{\mathcal{C} = \left\{X \mid X \in \mathcal{F} \text{ and } \nexists Y \supset X \text{ such that } sup(X) = sup(Y)\right\}}$$

X is closed if all supersets of X have strictly less support, that is, $sup(X) > sup(Y)$, for all $Y \supset X$.

The set of all closed frequent itemsets $\mathcal{C}$ is a condensed representation, as we can determine whether an itemset X is frequent, as well as the exact support of X using $\mathcal{C}$ alone.

Minimal Generators

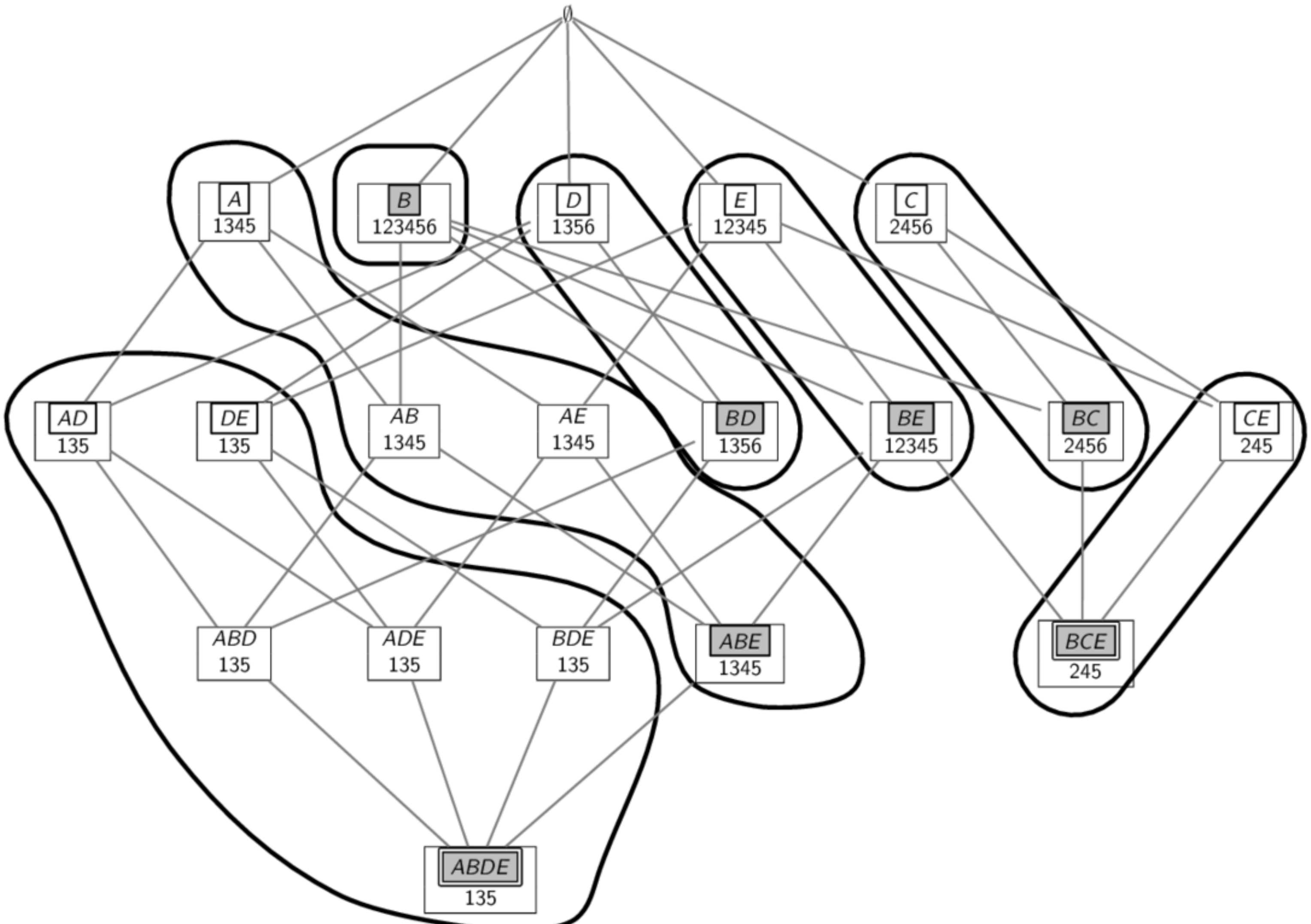
A frequent itemset X is a *minimal generator* if it has no subsets with the same support:

$$\mathcal{G} = \{X \mid X \in \mathcal{F} \text{ and } \nexists Y \subset X, \text{ such that } sup(X) = sup(Y)\}$$

In other words, all subsets of X have strictly higher support, that is, $sup(X) < sup(Y)$, for all $Y \subset X$.

Given an equivalence class of itemsets that have the same tidset, a closed itemset is the unique maximum element of the class, whereas the minimal generators are the minimal elements of the class.

Frequent Itemsets: Closed, Minimal Generators and Maximal



Itemsets boxed and shaded are closed, double boxed are maximal, and those boxed are minimal generators